

tmux // Terminal Multiplexer

Complete Technical Architecture, Control Algebra, Vi Buffers & Hardened Configuration

01. ARCHITECTURAL TOPOLOGY & PTY VIRTUALIZATION

Client-Server Execution Decoupling

tmux runs as an asynchronous event-driven daemon (``tmuxd``), detaching child processes from the originating GUI terminal or remote SSH session.



02. CONTROL MATRIX & KEYBINDING ALGEBRA

Session Lifecycle

GLOBAL

ACTION	CLI COMMAND	INTERACTIVE
New Session	<code>tmux new -s <name></code>	<code>:new -s <name></code>
Attach Target	<code>tmux attach -t <name></code>	—
Detach Client	<code>tmux detach</code>	Prefix <code>d</code>
Interactive Tree	<code>tmux choose-tree -s</code>	Prefix <code>s</code>
Rename Session	<code>tmux rename-session <n></code>	Prefix <code>\$</code>
Kill Session	<code>tmux kill-session -t <n></code>	<code>:kill-session</code>

Window Virtualization

TABS

ACTION	CLI / COMMAND	INTERACTIVE
Create Window	<code>tmux new-window -n <n></code>	Prefix <code>c</code>
Next / Previous	<code>tmux next-window / prev</code>	Prefix <code>n</code> / <code>p</code>
Direct Slot Jump	<code>tmux select-window -t <i></code>	Prefix <code>0..9</code>
Interactive Selector	<code>tmux choose-tree -w</code>	Prefix <code>w</code>
Rename Window	<code>tmux rename-window <n></code>	Prefix <code>,</code>
Terminate Window	<code>tmux kill-window</code>	Prefix <code>&</code>

Pane Geometry & Geometry Tiling

ERGONOMIC

OPERATION	DEFAULT BINDING	OPTIMIZED (.TMUX.CONF)	CORE SYSTEM COMMAND
Vertical Split (Side-by-Side)	Prefix %	Prefix	split-window -h -c "{pane_current_path}"
Horizontal Split (Stacked)	Prefix "	Prefix -	split-window -v -c "{pane_current_path}"
Directional Focus Navigation	Prefix Arrows	Prefix h j k l	select-pane -[U D L R]
Zoom / Fullscreen Toggle	Prefix z	Prefix z	resize-pane -Z (preserves layout state)
Cycle Tiling Layouts	Prefix Space	Prefix Space	next-layout (even-horiz, even-vert, tiled)
Synchronize Panes (Broadcast)	:setw synchronize-panes	Prefix y	Sends simultaneous keystrokes across all panes

03. THE VI BUFFER ENGINE & CLIPBOARD PIPELINE

Vi Copy Mode Pipeline

MEMORY BUFFER

Scrollback is stored in server-allocated memory, isolated from terminal viewport limitations.

1. Enter Mode: **Prefix** **[**
2. Vi Navigation: **h** **j** **k** **l**, **Ctrl+u**, **Ctrl+d**
3. Search: **/** (forward), **?** (backward)
4. Visual Select: **v**
5. Yank (Copy): **y** → passes to OS clipboard & buffer
6. Paste Buffer: **Prefix** **]**

Failure Remediation

DIAGNOSTICS

Vim Esc Delay	Set <code>escape-time 0</code> in config to remove modal delay.
Color Loss	Enforce <code>default-terminal "tmux-256color"</code> + RGB overrides.
Nested SSH	Press <code>Prefix</code> twice to pass hotkey to guest session.
Small Viewport	Enable <code>setw -g aggressive-resize on</code> .

04. HARDENED PRODUCTION CONFIGURATION (~/.TMUX.CONF)

```
# =====
# PhTex High-Performance Production .tmux.conf
# =====

unbind C-b
set -g prefix C-a                # Ergonomic prefix
bind C-a send-prefix

set -g default-terminal "tmux-256color"    # Full 24-bit Truecolor
set -ag terminal-overrides ",xterm-256color:RGB"
set -s escape-time 0                 # 0ms delay for Neovim/Vim
set -g history-limit 50000           # 50,000 line scrollback buffer
set -g mouse on                      # Full mouse support

set -g base-index 1                 # 1-based indexing for windows & panes
setw -g pane-base-index 1
set -g renumber-windows on

# Intuitive Split & Directory Preservation
bind | split-window -h -c "#{pane_current_path}"
bind - split-window -v -c "#{pane_current_path}"

# Vi Navigation Chords
bind h select-pane -L \; bind j select-pane -D \; bind k select-pane -U \; bind l select-pane -R
setw -g mode-keys vi
bind -T copy-mode-vi v send-keys -X begin-selection
bind -T copy-mode-vi y send-keys -X copy-pipe-and-cancel "wl-copy || xclip -in -selection clipboard"

# Minimalist Status Bar Design
set -g status-style "bg=#1e1e2e,fg=#cdd6f4"
set -g status-left "[bg=#89b4fa,fg=#11111b,bold]  #S #[bg=#1e1e2e,fg=#89b4fa] "
set -g window-status-current-format "[fg=#11111b,bg=#a6e3a1,bold] #I:#W "
bind r source-file ~/.tmux.conf \; display-message "Config Reloaded."
```